Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/30/2020
PI Name:	Ferri, James Ph.D.		
Project Title:	PASTA: Particle Stabilized Emulsions and Foams		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Virginia Commonwealth University		
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PI Web Page:			
City:	Richmond	**State:**	VA
Zip Code:	23219	**Congressional District:**	4
Comments:	NOTE: Previous affiliation was Lafayette College, Easton, PA until early 2018 when PI moved to Virginia Commonwealth University		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	06/01/2018	**End Date:**	05/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Tin, Padetha	**Contact Phone:**	216-433-8164
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0453		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE July 2018: Continuation of "Particle Stabilized Emulsions and Foams," grant NNX15AE98G, due to Principal Investigator move to Virginia Commonwealth University (VCU) in spring 2018. Overall aim of the PASTA (Particle Stabilized Emulsions and Foams) project is to develop a mechanistic description of the stabilization or destabilization of foams and emulsions, which can be accomplished by various combinations of surfactants, polymers, and particles. In this respect, we focus in particular on surface modification of (nano)particles by macromolecules (polymers) and biomacromolecules (proteins) and the concomitant changes to particle performance at the interface. Experimental work is contemplated on the three system levels -- the interface, the liquid film, and the formulated dispersion. Each level is interrogated so that structure-property relationships can be established along the hierarchy of increasing complexity. The comparison of foam and emulsion systems enabled through collaboration with the Foam Optics And Mechanics-Coarsening (FOAM-C) instrument, and the systematic consideration of surface modified (nano)particles as stabilizers/destabilizers will provide a new level of generalization. Experiments will be performed for the following hierarchies: 1. characterization of single interfaces with (nano)particles modified by polymers and proteins; 2. characterization of corresponding Gibbs (thin) and non-Gibbs (thick) emulsion films; 3. correlation between physico-chemical properties of (nano)particle modified interfaces and films, and; 4. correlation of (nano)particle physical chemistry, surface modification, interface and film properties with the behavior of disperse liquid-liquid systems (emulsions). In addition to the experimental program described above, theory will be developed which describes the behavior above in terms of the molecular and intrinsic constants associated with the materials of which they are composed. The composition of the consortium of participating teams (USA, Germany, Italy, Greece) is such that all aspects of the formulated problem are covered. The teams demonstrate a complementary expertise in form of the scientific fields and available experimental tools, as well as their capacity of tackling the theoretical side of the problems. Key experiments in microgravity conditions support the specific aims of this project. The basic processes involved in emulsions and foams, such as destabilization, mainly droplet aggregation and coalescence (destabilization of the film between droplets/bubbles), will be studied in the absence of segregation for model emulsions. In fact, during emulsion and foam destabilization, droplets larger than ten microns are rapidly segregating, making impossible on Earth the investigation of the above processes in controlled conditions. Microgravity experiments will then provide information about the features and characteristic times of these processes to be utilized to parameterize and validate the respective models. Since its inception, the PASTA project has sought to leverage the existing facilities on International Space Station (ISS) Fluid Science Laboratory (FSL). During the past three years, ESA (the European Space Agency) has refocused support in the Soft Matter Dynamics thrust to emphasize the FOAM-C facility which was originally developed to study coarsening and stability, respectively, of wet foams (multiple, polydisperse interfaces). As a result of shift in ESA priorities, in 2017 the PASTA team piloted the existing experimental module, FOAM-C, for suitability for emulsion dynamics studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Emulsions have large relevance for applications in a wide spectrum of technologies, processes, and products in different fields including oil and gas, specialty materials, pharmaceuticals, cosmetics, and foods. Both stable and unstable emulsions are needs that shared across a variety of applications. For example, efficient de-emulsifiers are needed to facilitate the process of phase separation (breaking) in the crude oil emulsions extracted from the wells, while high stability oil-in-water emulsions are required in the final stage of production of the same well or to facilitate pumping high-viscosity oils through pipelines. In the food industry, many products based on emulsified fats or oils (condiments, sauces, frozen confections, and others) require long term stability. However, destabilizing emulsions are key to processing raw food materials such as milk and vegetable oils.
Task Progress:	FISCAL YEAR 2019 REPORT COVERING JUNE 2018-MAY 2019 During 2018, we focused primarily on the transition of the PASTA/EDDI (Particle Stabilized Emulsions and Foams/Emulsion Dynamics and Drop Interfaces) project objectives to the experimental platform of the Soft Matter Dynamics facility (SMD) instrument on theory and modeling to translate experimental observations to physical processes. Technical discussions with collaborators in the EDDI network have identified the need to compare the diffusing wave spectroscopy (DWS) signal from the Aristotle University Thessaloniki -AUTH generator to the DWS data collected from systems produced in microfluidic emulsion generators in the VCU laboratories as a means to reduce (eliminate) polydispersity and characterize the experimental resolution of dynamic processes in liquid-liquid dispersions.
Bibliography Type:	Description: (Last Updated: 12/03/2024)
Articles in Peer-reviewed Journals	Ulaganathan V, Del Castillo L, Webber JL, Ho TTM, Ferri JK, Krasowska M, Beattie DA. "The influence of pH on the interfacial behaviour of Quillaja bark saponin at the air-solution interface." Colloids Surf B Biointerfaces. 2019 Apr 1;176:412-9. https://doi.org/10.1016/j.colsurfb.2019.01.017 ; PubMed PMID: 30665095 , Apr-2019
Articles in Peer-reviewed Journals	Goldhahn C, Schubert J, Schlaad H, Ferri JK, Fery A, Chanana M. "Synthesis of metal@protein@polymer nanoparticles with distinct interfacial and phase transfer behavior." Chemistry of Materials. 2018 Oct;30(!9): 6717-27. https://doi.org/10.1021/acs.chemmater.8b02314 , Oct-2018
Articles in Peer-reviewed Journals	Badruddozaa AZM, MacWilliams SV, Sebben DA, Krasowska M, Beattie D, Durian DJ, Ferri JK. "Diffusing wave spectroscopy (DWS) methods applied to double emulsions." Current Opinion in Colloid and Interface Science. 2018 Sep;37:74-87. https://doi.org/10.1016/j.cocis.2018.06.006 , Sep-2018

Articles in Peer-reviewed Journals	Krasowska M, Zawala J, Bradshaw-Hajek BH, Ferri JK, Beattie JK. "Interfacial characterisation for flotation: 1. Solid-liquid interface." Current Opinion in Colloid & Interface Science. 2018 Sep;37:61-73. https://doi.org/10.1016/j.cocis.2018.06.004 , Sep-2018
Articles in Peer-reviewed Journals	Pawliszak P, Zawala J, Ulaganathan V, Ferri JK, Beattie JK, Krasowska M. "Interfacial characterisation for flotation: 2. Air-water interface." Current Opinion in Colloid & Interface Science. 2018 Sep;37:115-27. https://doi.org/10.1016/j.cocis.2018.07.002 , Sep-2018